

Pick a NATS setup for two regions · v0.2

Every question in the guide, every answer, and what each answer adds to your setup. Use the HTML page to answer them one at a time. v0.2 supersedes v0.1. Each claim comes from the demo 03 lab report, measured 2026-09-28 and 2026-09-29 on one machine, nats-server 2.14.6. An open item marks what the lab did not settle.

Stage 1 · Accounts

Q1.1 Do you serve more than one customer?

A customer here is a separate business whose data must never be seen by another.

Yes · Give each customer its own account Each customer is walled off from the others.

- One account per customer. Answer the rest for one customer.

[Next: Q1.2 · Each region keeps its data private?](#)

No · One customer Carry on.

[Next: Q1.2 · Each region keeps its data private?](#)

Not sure Take the safer answer and add an open item.

- One account per customer, to be safe.
- Open item: Confirm whether more than one customer will use the system.

[Next: Q1.2 · Each region keeps its data private?](#)

Q1.2 Must each region keep its data private from the other region?

An account is a wall. Apps in one account cannot see another account's data unless you share it on purpose.

Yes · One account per region Each region owns its data.

- One account per region. Each region owns its data.

[Next: Q1.5 · One account reads another's data?](#)

No · One shared account Both regions see the same data.

- One shared account for both regions.

[Next: Q1.3 · Same stream name in both regions?](#)

Not sure Take the safer answer and add an open item.

- One account per region, to be safe.
- Open item: Confirm whether each region may see the other region's data.

[Next: Q1.5 · One account reads another's data?](#)

Q1.3 Must both regions make a stream with the same name?

A stream is a stored log of messages, such as all orders. In one shared account there is only one set of stream names.

Yes · Both regions need the same name This clashes. You will pick a way out next.

[Next: Q1.4 · Name clash: way out](#)

No · Each stream has one home region The other region reads it over the link.

- Each stream has one home region. The other region reads it over the link.

[Next: Q1.5 · One account reads another's data?](#)

Q1.4 These answers clash. Pick a way out.

In one shared account, two streams cannot have the same name. Over a gateway the second one is refused. Over a leaf link both are made, and if both listen to the same subjects every message is stored twice without a warning.

Switch to one account per region Each region makes its own stream with the same name.

- One account per region. Each region owns its data.

[Next: Q1.5 · One account reads another's data?](#)

Keep one account, and put the region in the name For example orders-east and orders-west.

- Stream names carry the region, for example orders-east and orders-west.

[Next: Q1.5 · One account reads another's data?](#)

Q1.5 Must one account read some of another account's data?

You can share chosen messages from one account to another. The other account gets new messages, not the stored history. NATS also documents a copy of a whole stream across accounts. That needs extra set-up, and this lab did not test it.

Yes · Share chosen messages on purpose Only what you pick crosses the wall.

- Share chosen messages from one account to another, on purpose.
- Give shared messages a prefix of their own. The prefix keeps them apart. It does not prove who sent them. Reserve it with publish permissions (E3a).

[Next: Q1.6 · Sharing both ways?](#)

No · Share nothing The wall holds.

- Accounts share nothing.

[Next: Q2.1 · Regions share any data?](#)

Q1.6 Must the sharing go both ways?

Yes · Share in each direction Set up sharing both ways.

- Sharing goes both ways.
- Open item: Two-way sharing between accounts. This was not tested in the lab.

Next: [Q2.1 · Regions share any data?](#)

No · Share one way Nothing flows back.

- Sharing goes one way only.

Next: [Q2.1 · Regions share any data?](#)

Stage 2 · Shape

Q2.1 Must the two regions share any data at all?

Yes · Link the regions The next questions pick the kind of link.

Next: [Q2.2 · Changes while a region is dark?](#)

No · No link Each region stands alone. You are done after this.

- Shape: T1 · No link. Each region stands alone.

Next: [your setup.](#)

Q2.2 If one region goes dark, must the other region still create or change streams?

Dark means down or cut off. Sending messages into a stream that already exists keeps working only while that stream's own servers still have a majority. A stream stored in the dark region stops. This question is only about making or changing streams.

Yes · Each region needs its own vote Changes need a majority of servers to agree.

Next: [Q2.3 · One name, same stream everywhere?](#)

No · A pause in changes is fine Use a gateway: one shared system.

- Shape: T2 · Gateway. Stream changes pause while a region is dark. Messages keep flowing into streams whose own servers are up.

Next: [Q3a.1 · Stream](#)

Not sure Take the safer answer and add an open item.

- Open item: Confirm whether stream changes must keep working while a region is dark.

Next: [Q2.3 · One name, same stream everywhere?](#)

Q2.3 Must one stream name mean the same stream in both regions?

With a shared system, orders is one stream wherever you ask for it. With separate systems, each region has its own orders. Separate systems can each store a live message, but that is not a backup: a region that was dark does not catch up on what it missed.

Yes · Keep one shared set of names The next question checks if that is possible.

[Next: Q2.4 · A third site?](#)

No · Each region keeps its own names Use a leaf link: each region runs on its own.

- Shape: T5 · Leaf link. Each region runs on its own.

[Next: Q3a.1 · Stream](#)

Q2.4 Is there a third site, with its own power and network?

With two equal regions, neither side has a majority alone. A small third site breaks the tie.

Yes · Use it as a tie-breaker Three servers there.

[Next: Q2.6 · Changes can wait on the third site?](#)

No · No third site You must give one thing up. You pick next.

[Next: Q2.5 · No third site: give up](#)

Q2.5 Without a third site, you cannot have both. Pick one to give up.

Two regions cannot both keep making changes during an outage and also share one set of names.

Give up changes during an outage Gateway. Names stay shared. Changes pause while a region is dark.

- Shape: T2 · Gateway. Stream changes pause while a region is dark. Messages keep flowing into streams whose own servers are up.

[Next: Q3a.1 · Stream](#)

Give up shared names Leaf link. Each region keeps working and keeps its own names.

- Shape: T5 · Leaf link. Each region runs on its own.

[Next: Q3a.1 · Stream](#)

Q2.6 Can every stream change wait for the link to the third site?

With a tie-breaker, each change needs a vote from across the link, so changes are only as fast as that link.

Yes · Use the tie-breaker site Either region can go dark and changes go on.

- Shape: T4 · Gateway with a tie-breaker site. Either region can go dark and changes go on. Streams stored in the dark region still stop (F28).
- Keep apps off the tie-breaker site. An app connected there stores its streams there.
- Open item: How long a change waits on the link to the third site. This was not tested in the lab.

[Next: Q3a.1 · Stream](#)

No · Changes cannot wait Use a leaf link: each region runs on its own.

- Shape: T5 · Leaf link. Each region runs on its own.
- Open item: Link speed between the regions. This was not tested in the lab.

[Next: Q3a.1 · Stream](#)

Stage 3a · Where streams live

Q3a.1 Name one stream you plan to have.

A stream is a stored log of messages, such as orders or vehicle positions. You answer the next questions once for each stream.

Type a stream name. Next: Q3a.2, asked once for each stream.

Q3a.2 Does only one region write to this stream?

Yes · Put it in that region Its home is where it is written.

- Lives in the region that writes it.

Next: Q3a.3 · Created from the other region?

No · Both regions write to it Make one stream per region.

- One stream per region, each written locally.
- Open item: this stream: each region's stream stops while its home region is dark.

Next: Q3a.3 · Created from the other region?

Q3a.3 Could the app that creates this stream be connected to the other region?

A stream is stored where the creating app is connected, unless you name the region.

Yes · Always name the region Otherwise it lands in the wrong region.

- Always name the region when you create it.

Next: Q3b.1 · Read while its home is dark?

No · The default is fine It lands next to the app.

- Created by an app in its home region. The default placement is fine.

Next: Q3b.1 · Read while its home is dark?

Not sure Take the safer answer and add an open item.

- Always name the region when you create it, to be safe.

Next: Q3b.1 · Read while its home is dark?

Q3b.1 Must the other region read this stream while its home region is dark?

A tie-breaker keeps changes working, but it does not bring back data stored in the dark region. Only a copy does.

Yes · Keep a copy in the other region The next questions set up the copy.

[Next: Q3b.2 · Reader in another account?](#)

No · No copy The other region reads it over the link while both are up.

- No copy. The other region reads it over the link while both regions are up.

[Next: Q3b.5 · Another stream?](#)

Not sure Take the safer answer and add an open item.

- Open item: this stream: confirm whether the other region must read it during an outage.

[Next: Q3b.2 · Reader in another account?](#)

Q3b.2 Is the reader in a different account from the one that holds this stream?

Yes · Share its messages The tested way. A copy that names only the stream copies the reader's own stream of that name, without an error.

- Share its messages to the reader's account. Do not copy it by stream name alone (E17, E18).
- Open item: this stream: shared messages stop while its home region is dark. The reader has no copy and no history.
- Open item: this stream: if the reader needs a real copy, NATS documents one across accounts: export the JetStream API, import it, and name it in the copy's external API setting. This was not tested in the lab.

[Next: Q3b.5 · Another stream?](#)

No · Keep a copy in the reader's region A read-only copy.

- A read-only copy in the reader's region.
- Only if the shape is T5: The copy crosses the leaf link. Tell it where to look on the other side, or it copies nothing without an error.

[Next: Q3b.3 · Copy up to date within a set time?](#)

Q3b.3 Must the copy of this stream be up to date within a set time?

Yes · Test it first How far a copy falls behind was not tested.

- Open item: this stream: how far the copy falls behind. This was not tested in the lab. Test it before you rely on it.

[Next: Q3b.4 · Copy takes writes during an outage?](#)

No · Some delay is fine The copy pauses while its source is dark.

- The copy may fall behind, and pauses while its source is dark.

[Next: Q3b.4 · Copy takes writes during an outage?](#)

Q3b.4 Must the copy of this stream take new data while its home region is dark?

Yes · Writes must go on This lab does not answer that. A copy is read-only.

- Open item: this stream: taking new data during an outage is not answered by this lab. A copy is read-only. This needs its own design.

[Next: Q3b.5 · Another stream?](#)

No · Read-only is enough Count its messages to prove it works.

- Count the copy's messages to prove it works. A wrong copy gives no error.

[Next: Q3b.5 · Another stream?](#)

Q3b.5 Do you have another stream to place?

Yes · Add another stream Answer the stream questions again.

[Next: Q3a.1 · Stream](#)

No · That is all of them Go on to link settings.

- Only if the shape is T5: One account spans both regions over the leaf link. Give each region's streams their own subjects.
- Only if the shape is T5: Open item: One account spans both regions over a leaf link. If a stream in each region listens to the same subjects, every message is stored twice without a warning. Give each stream its own subjects, or use one account per region.

[Next: link settings. T2 and T4 go to Q4.1, T5 to Q4.2, T1 to your setup.](#)

Stage 4 · Link settings

Q4.1 Will you give JetStream a domain name?

A domain is a name for one JetStream system. On a gateway, both regions are one system. A domain is not a security wall. Accounts are.

Yes · Use the same name on every server Different names split the regions and one side goes blind.

- The same JetStream domain name on every server.

[Next: your setup.](#)

No · Leave it unset Both regions share one system.

- No JetStream domain name. Both regions share one system.

[Next: your setup.](#)

Q4.2 Will system traffic cross the leaf link?

Your operator usually decides this. System traffic lets each side see the other side's servers.

Yes · Name a domain on each side The domain stops the two sides merging into one system. It is still not a security wall.

- A different JetStream domain name on each side of the leaf link. It keeps the systems apart. It is not a security wall.

[Next: Q4.3 · Reach the other side's streams by name?](#)

No · Nothing to set Each side stays separate.

- System traffic stays on its own side. Each side stays separate.

[Next: Q4.3 · Reach the other side's streams by name?](#)

Not sure Take the safer answer and add an open item.

- A different JetStream domain name on each side, to be safe.
- Open item: Ask the operator whether system traffic will cross the leaf link.

[Next: Q4.3 · Reach the other side's streams by name?](#)

Q4.3 Must one side reach the other side's streams by name?

A copy across a leaf link between different domains needs this. A copy across a gateway does not.

Yes · Name a domain on each side Then address the other side by its name.

- A domain name on each side. Address the other side by its name.

[Next: your setup.](#)

No · A domain is optional The sides stay apart anyway.

- No cross-side access by name is needed.

[Next: your setup.](#)

[For every setup](#)

Moving between T4 and T5

- Moving later between T4 and T5. The lab tested one way: stop every server, change the configs, start again on the same storage. In two runs (2026-09-28 and 2026-09-29) it has not shown safe, repeatable operation (D03-R11, D03-R12). T4 to T5 did not form separate vote groups either time; on 2026-09-29 the checked data was kept. T5 to T4 gave different results on the two days. With a stream name in both regions, some ZA copies were removed and the streams could not be read; whether they can be recovered, and why they were removed, is not known. Other ways, such as moving to newly built clusters, were not tested. It is not proven one-way. Choose before the data lands, and plan the move before you need it.

Rules for every setup

- Check the leader twice. After a failure, the status page can show an old leader for a while. Check again before you act.
- A named leader is not ready. The status page can name a new leader before changes work (F6a). Ready means a real change worked. Retry, and record the wait.
- Count the messages. A send can report success when nothing was stored.
- Check every copy by counting. A wrong or empty copy gives no error.
- A prefix is not proof of origin. It keeps imported messages apart. Reserve it with publish permissions.

Never pick these

- A single-server tie-breaker (T3). It leaves no room for one more failure.
- A gateway and a leaf link together (T6). It acts as a plain gateway.

The lab has not tested these yet

- A real network cut, with both regions alive but unable to reach each other.
- Loss of only some servers in a region.
- Any way to move between T4 and T5 other than the one tested. The tested way has not shown safe, repeatable operation.
- Why moving T5 to T4 removed ZA copies, and whether they can be recovered.
- A copy of a stream across accounts, set up as NATS documents it.
- The hub as the main store for both regions, under real load, and what its loss costs.
- How far a copy falls behind at production volume.

Sources

- [Lab report, generated by lab/run-all.sh](#)
- [Findings deck v1.0](#)
- [Pattern cards v0.4](#)
- [NATS docs: JetStream across accounts](#)
- [NATS docs: leaf nodes and JetStream domains](#)
- [NATS docs: leaf node remotes](#)